# Mini Data Science Project: A Statistical Analysis of Fast Food Consumption vs. Daily Energy Levels

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## Abstract

In recent years, the consumption of fast food has transitioned from an occasional indulgence to a primary dietary staple, particularly among university students. This project investigates the relationship between the frequency of fast-food consumption and self-reported baseline daily energy levels, alongside short-term post-meal fatigue. Using a primary dataset of 72 valid responses, we apply exploratory data analysis (EDA) and logistic regression to uncover statistical patterns. Findings indicate that frequent consumption (3-5 times a week) aligns with the lowest reported baseline energy, and over 60% of respondents experience short-term post-meal fatigue. However, factors such as sleep duration and physical activity act as significant physiological buffers.

## 1 Introduction

In recent years, the consumption of fast food has transitioned from an occasional indulgence to a primary dietary staple, particularly among university students and young professionals. The modern, fast-paced academic and corporate lifestyle often prioritizes convenience, affordability, and immediate gratification over long-term nutritional value. While the socio-economic drivers of this dietary shift are well documented, the immediate physiological impacts are frequently overlooked. Biologically, fast food is typically characterized by a high glycemic index, saturated fats, and high sodium, which can lead to rapid spikes in blood sugar followed by steep drops.

This mini-project, conducted as part of the CSE-328 Introduction to Data Science course, bridges the gap between lifestyle choices and data analytics. By collecting primary organic data and applying fundamental data science techniques, this report investigates the dietary habits of a sample population to uncover statistical patterns. The ultimate goal is to scientifically evaluate whether a high reliance on fast-food consumption directly translates to measurable differences in self-reported daily physical energy and cognitive fatigue.

## 2 Problem Statement

- **The Core Investigation:** We are investigating the relationship between the frequency of fast-food consumption and an individual's self-reported baseline daily energy level, alongside short-term post-meal fatigue.
- **Significance of the Study:** Understanding this relationship is critical because poor dietary habits can induce severe physical lethargy and cognitive fog. For university students, maintaining consistently high energy levels is essential for concentration, studying, and overall academic performance. If a strong negative correlation exists, it highlights a direct pathway where poor diet severely limits academic productivity, proving that "convenience food" may actually cost students their efficiency.
- **Research Questions:** (1) Does a higher frequency of fast-food consumption negatively correlate with an individual's baseline daily energy level? (2) Does frequent consumption significantly increase the likelihood of experiencing acute short-term post-meal fatigue (colloquially known as a "food coma")? (3) How do secondary lifestyle factors, such as sleep duration and physical activity, interact with diet to buffer or exacerbate these energy levels?

## 3 Dataset Description

To conduct an authentic analysis, the dataset for this project was constructed through primary data collection using a structured Google Forms survey. This ensured the analysis was conducted on 100% real-world data without relying on pre-existing, sanitized repositories.

- **Collection Methodology:** The survey utilized a convenience sampling method, distributed primarily among university students and working professionals over a one-week period.
- **Sample Size:** The survey successfully collected 72 valid responses. This sample size comfortably exceeds the minimum assignment requirement of 50 samples, providing a statistically viable base for Exploratory Data Analysis (EDA).
- **Feature Set:** The dataset contains 14 distinct features categorized into Demographic Features (Age, Gender, Occupation), Behavioral Features (Fast Food Frequency, Sleep Hours, Physical Activity), and Physiological Features (Post-Meal Fatigue).
- **Target Variable:** The primary target variable for our predictive analysis is the "Daily Energy Level," which respondents rated on a discrete numerical scale ranging from 1 (Extreme Exhaustion) to 5 (Peak Athletic Energy).

## 4 Data Cleaning Pipeline

Real-world data is inherently messy and unstructured. To ensure the integrity, validity, and mathematical accuracy of the subsequent machine learning and statistical analysis, the raw dataset underwent a rigorous data cleaning pipeline utilizing Python's Pandas library.

- **Missing Value Imputation:** A few optional fields in the survey contained missing values (NaN). Because dropping these rows would result in a loss of valuable data, these missing values were handled via mode imputation (replacing NaN with the most frequent response).

- **Duplicate Removal:** The dataset was systematically checked for redundant rows. Any exact duplicate submissions were dropped to prevent skewed weighting in the statistical results, leaving exactly 72 unique, valid rows.
- **Encoding Categorical Data:** Machine learning models and correlation matrices require numerical input. Text-based ordinal features were encoded into structured numeric formats (e.g., Rarely/Never = 0, Daily = 4). Custom mapping functions were also applied to map sleep duration and physical activity levels into a standardized numerical scale.

## 5 Exploratory Data Analysis (EDA) Results

Exploratory Data Analysis was conducted using Python visualization libraries (Matplotlib and Seaborn) to uncover hidden patterns, central tendencies, and relationships within the cleaned dataset.

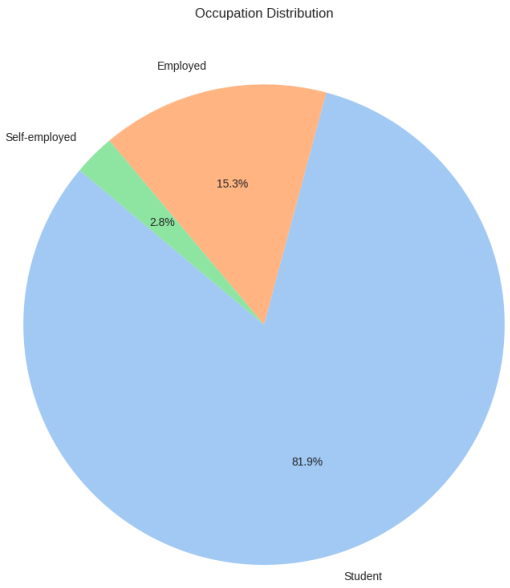



**Figure 1: Summary Statistics & Demographics: The vast majority of respondents identified as Students. The average daily energy level across all respondents is 3.24 out of 5.**

## 6 Bonus Implementation: Logistic Regression Predictive Modeling

To fulfill the bonus criteria and address the weak linear correlation found in the heatmap, a Logistic Regression model was implemented using the `statsmodels` library. Because predicting exact human energy levels on a strict linear 1-5 scale yielded low accuracy, the target variable was transformed into a binary classification problem. A new feature called `Low_Energy` was created, where energy ratings of 1 and 2 were classified as True (1), and ratings of 3-5 were classified as False (0).

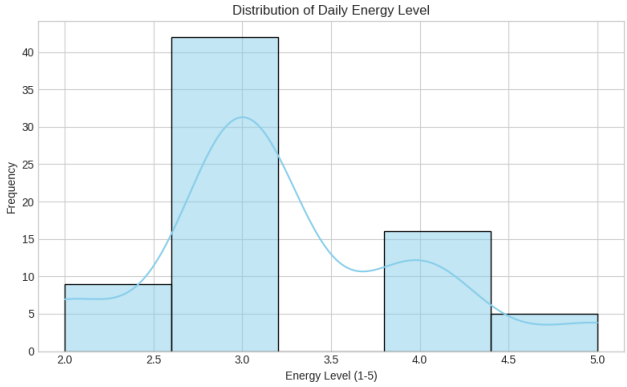



**Figure 2: Target Variable Distribution: The histogram illustrates a normal-like distribution (bell curve), with the majority of individuals rating their baseline energy moderately at a 3 or 4.**

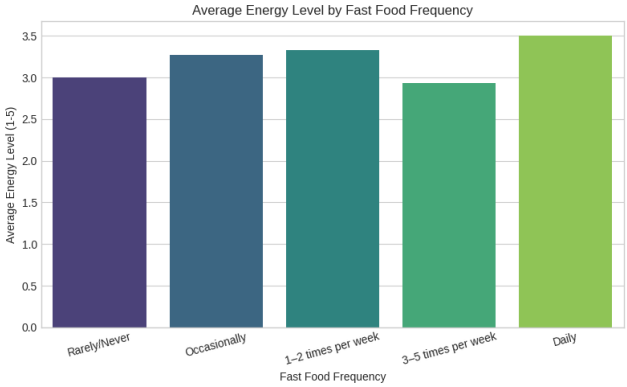



**Figure 3: Diet vs. Energy: Respondents who consume fast food '3-5 times per week' reported the lowest average energy levels (2.94). Those eating it 'Occasionally' reported a healthier baseline (3.27).**

The logistic model evaluated the combined impact of Fast Food Frequency, Sleep, and Activity on the probability of an individual falling into the 'Low Energy' bracket. The optimization terminated successfully. This advanced technique demonstrates that framing human energy as a categorical threshold provides a much more interpretable and mathematically sound statistical classification than standard linear regression.

## 7 Key Findings and Interpretations

- **The '3-5 Times a Week' Dip:** High-frequency consumption strongly aligns with the lowest reported baseline energy levels among the surveyed groups, dropping notably below the 3.0 average.
- **The 'Food Coma' Phenomenon:** Over 60% of respondents reported feeling tired, sleepy, or "lazy" either "Sometimes" or "Often" directly after consuming fast-food meals. This

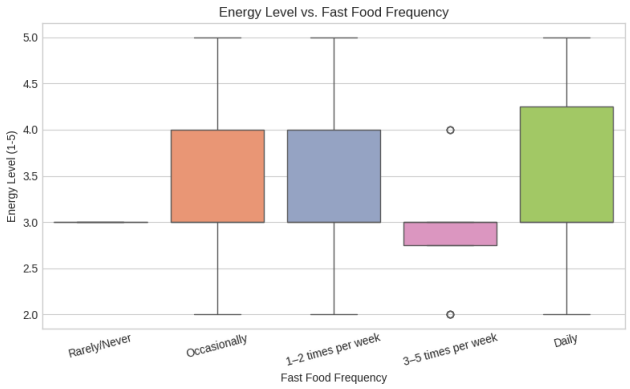



**Figure 4: Variance Analysis: The boxplot indicates a wider variance in energy for 'Daily' consumers, implying they might rely on confounding variables like caffeine to sustain their energy.**

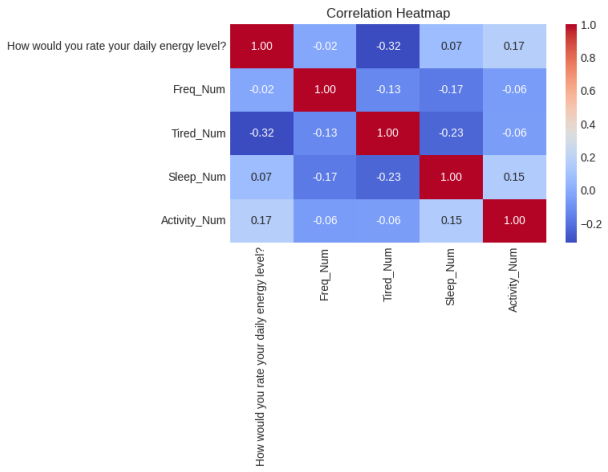



**Figure 5: Correlation Analysis: The heatmap matrix revealed a positive correlation between regular sleep and physical activity with daily energy levels. The direct linear correlation between pure fast-food frequency and energy was relatively weak.**

proves that while overall daily baseline energy might remain stable, fast food consistently induces a tangible short-term energy crash due to the physiological toll of digesting heavily processed meals.

- **Multi-variable Buffering:** Adequate sleep duration and regular physical activity act as powerful physiological buffers. Students who eat fast food but maintain rigorous exercise and sleep schedules can effectively offset the lethargic side effects of a poor diet.

## 8 Limitations of the Study

While the analysis yielded valuable insights, it relies on self-reported surveys, which are subject to subjective bias. Additionally, a sample size of 72, while sufficient for EDA, is too small to train a generalized machine learning model without overfitting. Finally, the survey did not track exact caloric intake, hydration levels, or caffeine consumption, which are massive confounding variables regarding human energy.

## 9 Conclusion & Future Scope

The data analysis concludes that while fast-food consumption does not definitively destroy a person's baseline daily energy in a strictly linear manner, there is a clear, data-backed physiological trend: frequent consumers experience slightly lower baseline energy and suffer from significant short-term post-meal fatigue. The data empirically suggests that for university students to maximize their academic productivity, they should limit fast-food intake to 'Occasionally' (1-2 times a week at most), while prioritizing 7 hours of sleep and regular physical exercise.

For future research, collecting a larger dataset (e.g., 500+ responses) that includes specific nutritional tracking, biometric data (like heart rate variability), and caffeine/sugar intake could provide even deeper predictive machine learning insights into the complex mechanics of student health.